

Dhruv Bamal

BACKEND ENGINEER

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SUMMARY

Backend engineer with a systems-design foundation and 200+ DSA problems solved in C++. Builds production-grade APIs, databases, and distributed systems using Node.js, TypeScript, PostgreSQL, and Redis. Focused on correctness under concurrency: ACID transactions, row-level locking, idempotency, state machines, and queue-backed background processing, containerised with Docker.

SKILLS

Languages: TypeScript, JavaScript, C++, Java, SQL

Backend & APIs: Node.js, NestJS, Express.js, REST API design, JWT, OAuth, Swagger / OpenAPI

Databases: PostgreSQL, SQL, Prisma ORM, transaction safety, row-level locking, indexing

Reliability: ACID transactions, idempotency, retry strategies, exponential backoff, state machines, concurrency control

Distributed & Async: Redis, BullMQ, asynchronous queues, delayed jobs, event-driven architecture, multi-tenancy, RBAC, fault tolerance

DevOps, Testing & Web: Docker, Docker Compose, CI/CD (GitHub Actions), Jest, Supertest, k6, Git, React 18, Next.js

PROJECTS

ProcureFlow — Multi-Tenant Purchase and Expense Approval API

Stack: NestJS, TypeScript, PostgreSQL, Prisma, Redis, BullMQ, Docker Compose, GitHub Actions

- Built a multi-tenant approval API that routes purchase requests and expense claims through role- and amount-based approval rules across organizations, branches, and departments.
- Implemented JWT authentication with hashed, rotated refresh tokens and RBAC guards over five roles; tenant ownership is verified server-side on every protected query rather than trusted from client input.
- Recorded every workflow decision — approved, rejected, or returned for changes — in an immutable audit log, offloaded monthly CSV reports to BullMQ workers, and covered cross-tenant access and approval transitions with Jest/Supertest tests in CI.

SlotSure — Capacity-Aware Appointment and Reservation API

Stack: NestJS, TypeScript, PostgreSQL (with raw SQL for lock semantics), Prisma, Redis, BullMQ, k6

- Designed an atomic reservation algorithm — validate input, lock the slot-capacity row, verify remaining capacity, create a held reservation with expiry, decrement capacity — committed in a single transaction to prevent double-booking.
- Guaranteed exactly-once capacity release: expiry workers are idempotent, so a repeated job cannot restore a seat twice, and idempotency keys stop client retries from creating a second reservation.
- Enforced a reservation state machine (held to confirmed, expired, or cancelled) rejecting invalid transitions, and proved it with concurrency tests where two simultaneous requests race for the final slot and exactly one succeeds.

DealerSync — Asynchronous Order-Import and Exception-Management API

Stack: NestJS, TypeScript, PostgreSQL, Redis, BullMQ, streaming CSV parser, Docker Compose

- Built an asynchronous CSV order-import pipeline in which the upload request returns immediately and a dedicated worker parses rows in bounded batches, keeping request latency independent of file size.
- Implemented independent per-row validation with partial success, so invalid rows are recorded with actionable reasons and downloadable error reports without discarding valid orders.
- Made workers idempotent using deterministic keys and unique database constraints, so retries with exponential backoff cannot create duplicate orders; import progress and status persist across worker restarts.

EXPERIENCE

Y Combinator Startup Simulation — Forage Virtual Internship

- Completed Y Combinator's working-startup simulation, covering early-stage company dynamics, product-market fit, and entrepreneurial problem-solving.

EDUCATION

B.Tech, Computer Science and Engineering

SRM Institute of Science and Technology, Ghaziabad | 2023 – 2027 | CGPA: 7.35 / 10

ACHIEVEMENTS

- 1st Place, Prompt Rachna 2.0 (Techstacy 2.0 Hackathon)** — competing as team Spring Storm. Qualified through the virtual round and won the live finals at Unstop Headquarters, Saket, Delhi, among 100+ participants. Built Helios Protocol, a branching interactive storytelling web experience with multiple endings, quick-time events, and integrated mini-games. Live demo: [cyber-death2-0.vercel.app](#)
- 200+ problems solved on LeetCode**, with consistent practice in C++.